

Unit 14 - Equations and Inequalities

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The following B.E.S.T Standards will be covered in this unit:

MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

- *Clarification 1: Variables may be on the left or right side of the inequality symbol.*

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

- *Clarification 1: Instruction includes using manipulatives, drawings, number lines and inverse operations.*
- *Clarification 2: Instruction includes equations in the forms $x + p = q$ and $p + x = q$, where x, p and q are any integer.*
- *Clarification 3: Problems include equations where the variable may be on either side of the equal sign.*

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

- *Clarification 1: Instruction includes using manipulatives, drawings, number lines and inverse operations.*
- *Clarification 2: Instruction includes equations in the forms $x/p = q$, where $p \neq 0$, and $px = q$.*
- *Clarification 3: Problems include equations where the variable may be on either side of the equal sign.*

MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.

- *Clarification 1: Instruction focuses on using algebraic reasoning, drawings, and mental math to determine unknowns.*
- *Clarification 2: Problems include the unknown and different operations on either side of the equal sign. All terms and solutions are limited to positive rational numbers.*

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

- *Clarification 1: Instruction focuses on the properties of inequality. Refer to Properties of Operations, Equality and Inequality (Appendix D).*
- *Clarification 2: Instruction includes inequalities in the forms $px > q$; $x/p > q$; $x \pm p > q$ and $p \pm x > q$, where p and q are specific rational numbers and any inequality symbol can be represented.*
- *Clarification 3: Problems include inequalities where the variable may be on either side of the inequality symbol.*

Unit 14, Lesson 1: Translating Real-World Written Descriptions into Algebraic Inequalities



MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.



- ☐ I can translate a written description into an algebraic inequality.
- ☐ I can represent an inequality on a number line when given a written description of a real-world context.



In the U.S., the average LCD TV screen has a width of 3.63 feet (ft) and a height of 2.04 ft.

Doak Campbell stadium has a jumbotron screen with a width of 119.7 ft and a height of 56.7 ft.

1. Estimate how many TV screens can be laid on top of the jumbotron at Doak Campbell stadium.
2. Explain how you found your estimate.

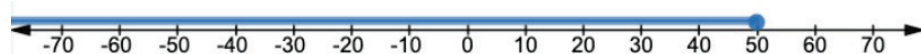




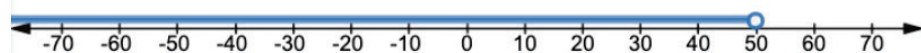
14.1.2 Exploration: Representing Inequalities on a Number Line

1. Complete the following problems.

- Choose an x -value that will make the inequality $x \leq 50$ true.
- Choose an x -value that will make the inequality $x \leq 50$ false.
- Have your partner verify your choices are correct.
- What is the x -value that is true for $x \leq 50$ but false for $x < 50$?
- The number line below represents all of the x -values that are true for $x \leq 50$.



The number line below represents all of the x -values that are true for $x < 50$.

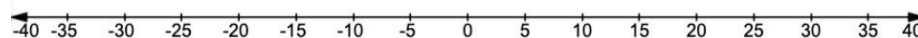


How is the graph for $x < 50$ different from the graph for $x \leq 50$?

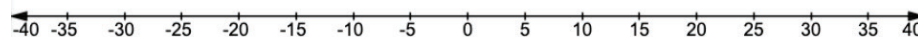
2. Complete the following.

- How are the solutions to $x \geq -23$ and $x > -23$ the same?
- How are the solutions to $x \geq -23$ and $x > -23$ different?
- Graph the solutions to each inequality on the number lines.

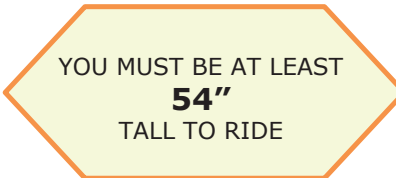
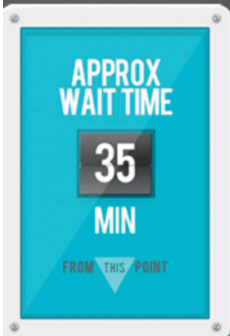
$$x \geq -23$$



$$x > -23$$



3. Complete the following.
- a. With your partner, determine which inequality best represents each context.

Nutritional Information <table><tr><td>Calories:</td><td>2,000</td></tr></table> <table><tr><td>Total Fat</td><td>Less than</td><td>65g</td></tr><tr><td>Sat Fat</td><td>Less than</td><td>20g</td></tr><tr><td>Cholesterol</td><td>Less than</td><td>300mg</td></tr><tr><td>Sodium</td><td>Less than</td><td>2,400mg</td></tr></table> Total Fat $f \leq 65$ $f \geq 65$ $f < 65$ $f > 65$	Calories:	2,000	Total Fat	Less than	65g	Sat Fat	Less than	20g	Cholesterol	Less than	300mg	Sodium	Less than	2,400mg	 $h \leq 54$ $h \geq 54$ $h < 54$ $h > 54$
Calories:	2,000														
Total Fat	Less than	65g													
Sat Fat	Less than	20g													
Cholesterol	Less than	300mg													
Sodium	Less than	2,400mg													
 $p \leq 10$ $p \geq 10$ $p < 10$ $p > 10$	 $s \leq 50$ $s \geq 50$ $s < 50$ $s > 50$														
 $x \leq 50$ $x \geq 50$ $x < 50$ $x > 50$	 $w \leq 35$ $w \geq 35$ $w < 35$ $w > 35$														

- b. In which, if any, of these situations does a value of 0 make sense given the context?
- c. In which, if any, of these situations does a negative number make sense given the context?
- d. In which, if any, of these situations does 1,000 make sense given the context?

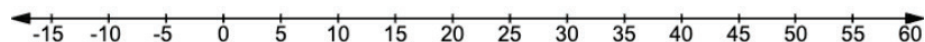


14.1.3 Guided Practice: Translating a Written Description into an Algebraic Inequality

1. According to the Interstate Highway Standards, U.S. and state highway traffic lanes must be at least 15 feet wide.

a. Write an inequality to represent the width, w , that traffic lanes can be.

b. Represent the inequality on the number line.



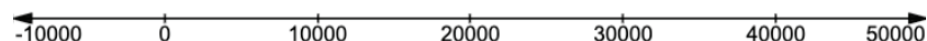
c. List two examples of possible widths, in feet, for the traffic lane.

d. Give one example of a width that is true for the inequality but does not make sense in the real world.

2. High-altitude parachute jumps are usually made for an altitude that is no more than 35,000 feet to avoid frostbite from extreme cold.

a. If h represents the height of a jumper, write an inequality to describe all possible values of h .

b. Represent the inequality on the number line.



c. Does the inequality that represents the situation include 0 as a possible height?

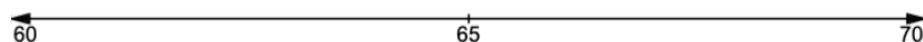
d. Explain why a negative number does not make sense for this context.



14.1.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

1. The inequality $67.85 \leq m$ represents the amount of money, m , Edin needs to earn to buy a new bike.

a. Graph the inequality.



b. Give one value that is true for $67.85 \leq m$ but is unreasonable for the context.

c. Explain why the value you chose is unreasonable.

2. A farmer plans to plant more than 52.75 acres of oranges.

a. If A represents the number of acres of oranges the farmer plants, write an inequality that represents the number of acres the farmer plants.

b. Circle all the possible values for A , the number of acres the farmer plans to plant, from the list below.

52.7 acres

$52\frac{3}{4}$ acres

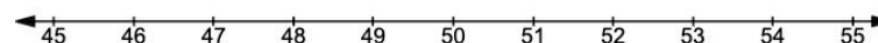
53 acres

52 acres

52.8 acres

$52\frac{1}{4}$ acres

c. Graph the inequality.





14.1.5 Your Turn!

1. Would you rather drive a go-cart with speed, g , in miles per hour (mph), represented by the inequality $12.5 \geq g$, or an electric bike with speed, b , in mph, represented by $b \geq 7\frac{1}{2}$?

a. I choose the ☐ go-cart ☐ electric bike because _____

_____.

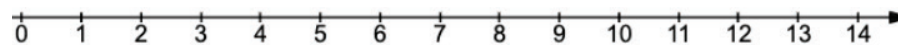
The ☐ go-cart ☐ electric bike could have a speed of 18 mph.

- b. Circle all the values from the list below that are possible speeds for the go-cart.

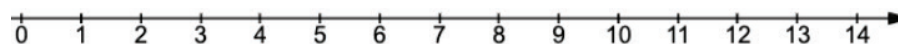
1.25 mph 15.2 mph $10\frac{1}{3}$ mph $\frac{12}{5}$ mph

12.45 mph 11.95 mph $12\frac{2}{3}$ mph $11\frac{11}{12}$ mph

- c. Represent the speed of the electric bike on the number line.



- d. Represent the speed of the go-cart on the number line.



- e. Explain why the number lines do not include negative numbers.



14.1.6 Wrap-Up: Ticket Out the Door

At Universal Studios, a person must be at least 48 inches (in.) tall to ride Revenge of the Mummy™.

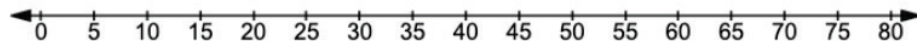
1. Write an inequality representing the height, h , a person must be to go on the ride.
2. Circle all values from the list below that satisfy this inequality.

47.9 in. 100 in. $\frac{48}{2}$ in. 48.5 in.

$70\frac{3}{4}$ in. $48\frac{1}{4}$ in. 0 in. 400 in.

48.099 in. 48 in. $50\frac{3}{4}$ in. 47.9 in.

3. Graph the inequality on the number line.



Unit 14, Lesson 2: Finding an Unknown Value in an Equation



Benchmark

MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.



Learning Targets

- ☐ I can determine an unknown decimal in a numeric equation with three numbers.
- ☐ I can determine an unknown fraction in a numeric equation with three numbers.



14.2.1 Warm-Up: What is STEM?

Science, Technology, Engineering, and Mathematics (STEM) is a way of thinking and doing. STEM skills are useful to almost every career.

An individual does not need to be a master in all four areas of STEM to be successful; teamwork plays an important role. Playing to a person's strengths and working as a team creates a greater impact than working alone.

1. Describe a time where it was better for you to work as a team than alone.
2. List two jobs that involve working as a team but are in different fields of STEM.



14.2.2 Refresher: The Unknown Value

With your partner, complete the following.

In math, **variables** are used to represent unknown values.

In an algebraic expression, a **variable** is a letter that represents a number.

1. Circle the variable(s) in each expression.

$$m + 5$$

$$2x + y$$

$$\frac{a}{4}$$

Variables are found in algebraic equations.



An **equation** is a mathematical relation statement where two equivalent expressions or values are separated by an equal sign.

An **algebraic equation** is an equation with at least one variable.

2. Circle the algebraic equations.

$$6 + 8 = 14$$

$$5y = 35$$

$$2w + 7 = 13$$

$$\frac{12}{3} + 8 = 12$$



To **solve** an equation means to find the solution.

The **solution(s)** to an algebraic equation is (are) the value(s) of the variable(s) that makes the equation true.



14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

For some equations, the solution can be found mentally or by using reasoning.

1. The solution to the equation $n - 12.1 = 35.3$ can be found using reasoning.
 - a. Discuss with your partner how to find the value of n in the equation $n - 12.1 = 35.3$.
 - b. The equation is rewritten vertically with a box in the place of the variable. Write the solution in the box and check whether it makes the equation true.

$$\begin{array}{r} \boxed{} \\ - 12.1 \\ \hline 35.3 \end{array}$$

Diagrams can also be helpful to find a solution.

2. Olivia gave equal amounts of money to each of her 3 children. Olivia gave each child \$16.75.

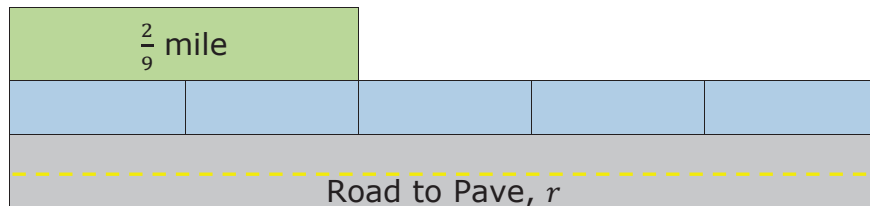
The bar model represents the equation $\frac{m}{3} = 16.75$, where m is the total amount of money Olivia gave all of her children.

16.75	16.75	16.75
m		

- a. How can the bar model be used to find the total amount of money Olivia gave all of her children, m ?
- b. What number can be divided by 3 to get a result of 16.75?

$$\frac{\boxed{}}{3} = 16.75$$

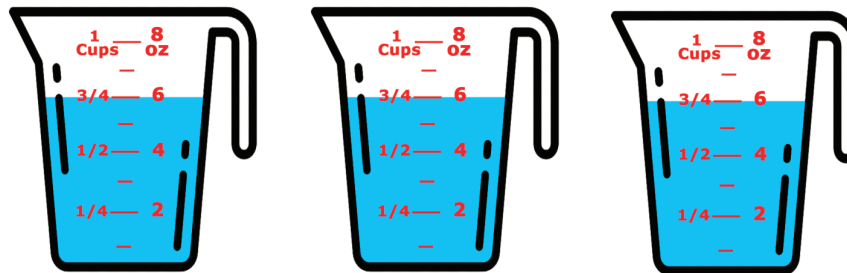
3. A road construction crew paved $\frac{2}{9}$ mile, which is $\frac{2}{5}$ of a full paving job. The bar model represents the equation $\frac{2}{9} = \frac{2}{5}r$, where r is the length of the whole road being paved.



- Discuss with your partner how the diagram represents the information given.
- Describe how to use the bar model to solve the equation for r , the length of the road being paved.
- How long is the road being paved?

d. Check your solution. $\frac{2}{9} = \frac{2}{5} \left(\frac{\quad}{\quad} \right)$

4. Donté used milk for 3 batches of a recipe. He used $\frac{3}{4}$ cup of milk for each batch.



The equation $m \div 3 = \frac{3}{4}$ can be used to find m , the amount of milk Donté used.

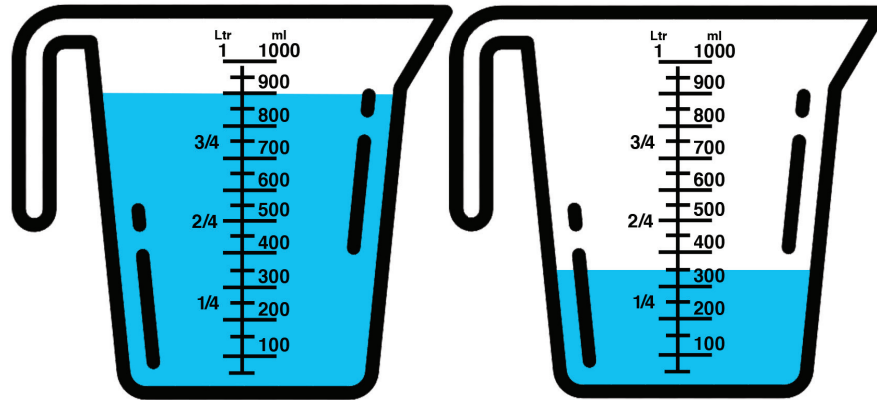
- Use the measuring cups shown to solve the equation $m \div 3 = \frac{3}{4}$.
- Check your solution. $\frac{\quad}{\quad} \div 3 = \frac{3}{4}$



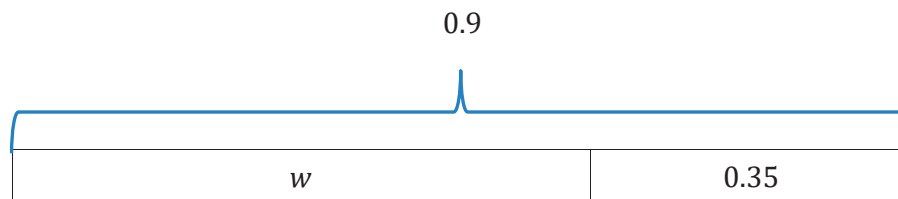
14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams

With your partner, solve the following problems using reasoning and diagrams.

- Vincent had 0.9 liters (L) of water. He poured some into the mixing bowl and had 0.35 L left.



The bar model shown represents this situation.



- Discuss with your partner how the bar model represents Vincent's situation. Then, write a brief summary of your discussion.

- The equation $0.9 - w = 0.35$ can also be used to represent this situation. Find the value of w , the amount of water Vincent poured into the mixing bowl.

- Josephine told her brother she would give him $\frac{1}{2}$ of the money they made from hosting a lemonade stand at their school carnival. She gave her brother \$24.25. This situation can be represented by the equation $\frac{1}{2}x = 24.25$ to find x , the amount of money they made at the lemonade stand.

- Use the bar model shown to represent the situation.

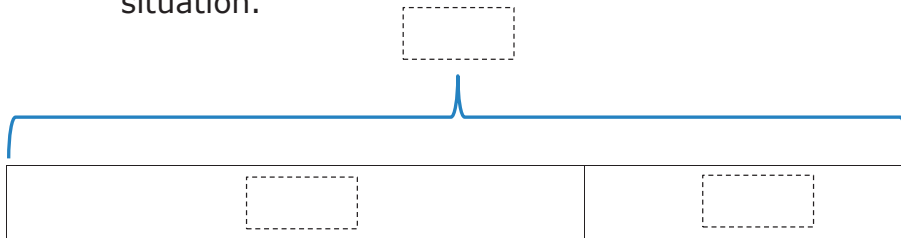
- Find the value of x , the amount of money they made at the lemonade stand.



14.2.5 Your Turn!

1. DeAndré paid \$58.54 for a game at Store A, which was \$9.21 less than the price of the game at Store B. This situation can be represented by the equation $58.54 = b - 9.21$, where b is the price of the game at Store B.

- a. Complete the bar model diagram to represent this situation.



- b. Find the value of b .

2. Mr. Bing placed a new 5-gallon water bottle in the office water dispenser when he arrived in the morning. At the end of the day, the bottle had $2\frac{7}{8}$ gallons remaining. The equation $5 = x + 2\frac{7}{8}$ can be used to find x , the gallons of water dispensed from the bottle.

- a. Draw a diagram representing the situation.

- b. How much water was dispensed from the bottle? Include units.



14.2.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered...	
One mistake I learned from today...	One thing I am not sure about...YET!
One thing I am confident I know is...	



Unit 14, Lesson 3: Discovering Inverse Operations



Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



Learning Targets

- ☐ I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations.
- ☐ I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.



Dale is helping to fix the roof at his uncle's house. They are using a 40-foot ladder. Dale is climbing the ladder to reach the roof. He is currently 7 feet up the ladder.

1. Sketch a picture to represent this situation.



Algebra tiles can be used to represent and solve algebraic equations.

In the model shown, **1** represents 1 and **x** represents x .

1. The steps to determine the value of x using algebra tiles are shown.
 - a. Work with your partner to complete the last step according to the description.

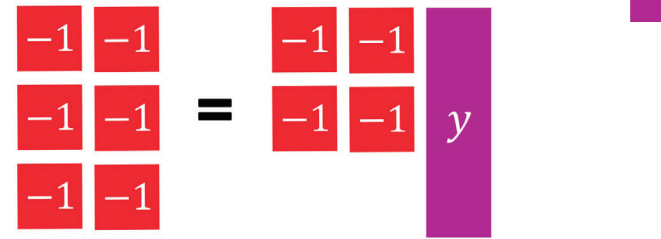
Algebra Tiles	Description
	Use algebra tiles to represent the equation $x + 3 = 5$.
	Determine any tiles that are represented on both sides of the equation.
	Remove the equal parts from both sides of the equation.

- b. The expressions $x + 3$ and 5 are equivalent according to the given equation. Discuss with your partner whether the tiles you drew on each side of the equation mat in the last step represent an equivalent relationship. Then, write a brief summary of your discussion.

- c. How many blue tiles are equivalent to one teal tile?

- d. What is the value of x in the equation $x + 3 = 5$?

2. In the model shown, -1 represents -1 and represents y .



- a. Work with your partner to determine how many red tiles are equivalent to one purple tile.

- b. What is the value of y in the equation $-6 = -4 + y$?



14.3.3 Guided Instruction: Solving Equations Using Inverse Operations

An equation relates two equivalent expressions. When solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. Discuss with your partner what it means for an equation to remain balanced. Then, summarize your discussion.

When finding the solution to an equation, inverse operations are used to balance an equation.

An **inverse operation** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

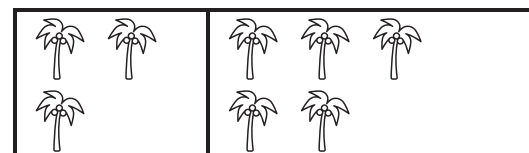
Operation	Inverse Operation
Subtraction (−)	
Addition (+)	



14.3.4 Guided Instruction: Solving Equations With Inverse Operations

Bar models and balance scales are useful tools for representing equations.

1. In the bar model shown, each palm tree represents 1 unit.

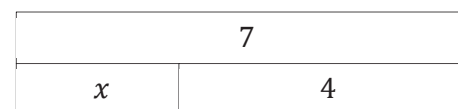


The bar model can be used to represent either an addition or subtraction equation. Fill in the missing values in each equation to match the bar model.

$$3 + \underline{\hspace{1cm}} = 8$$

$$8 - 3 = \underline{\hspace{1cm}}$$

2. The bar model shown can be used to represent either an addition or subtraction equation.



- a. Circle the addition equation that matches the diagram.

$$x + 7 = 4$$

$$7 = x + 4$$

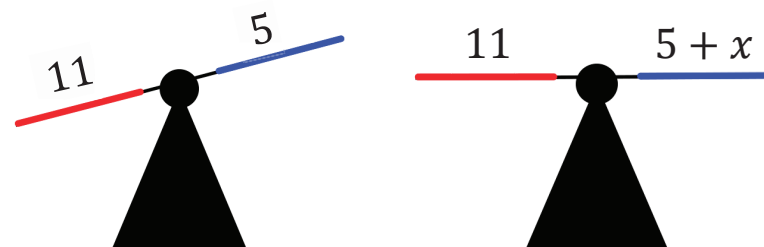
$$x = 7 + 4$$

b. Write an equivalent equation using the inverse operation.

c. Justify to your partner that the equation you wrote is an equivalent equation.

d. What is the value of the variable, x ?

3. Two balance scales are shown. The scale on the left is unbalanced. To balance the scale, an unknown value, x , needs to be added to the right side of the scale.



a. The equation represented by the balance scale on the right is shown. Use inverse operations to find the value of x by completing the steps shown.

$$x + 5 = 11$$

$$x + 5 - \square = 11 - \square$$

$$x = \square$$

b. Rewrite the original equation, substituting the solution for x .

c. Explain how to use the equation in Part B to confirm that your solution is correct.

Inverse operations can be used to find the value of the variable in an algebraic equation by creating equivalent equations.



Addition Property of Equality

If $a = b$, then $a + c = b + c$

Subtraction Property of Equality

If $a = b$, then $a - c = b - c$

4. The equation $x + 23 = 78$ can be solved by using inverse operations.

- a. Fill in values in each equivalent equation to solve for x .

$$x + 23 = 78$$

$$x + 23 - \underline{\hspace{2cm}} = 78 - \underline{\hspace{2cm}}$$

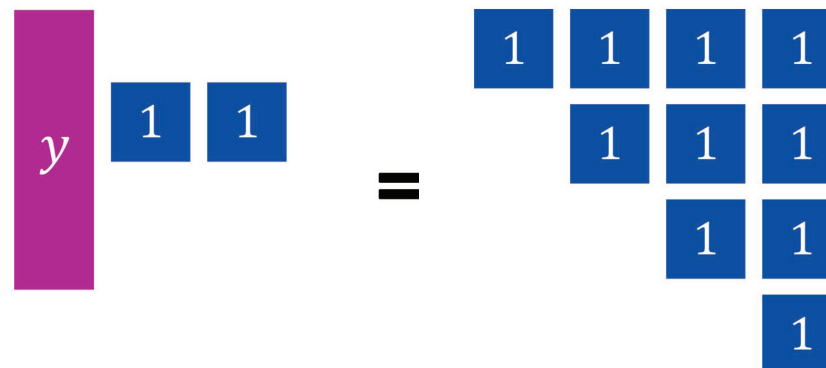
$$x = \underline{\hspace{2cm}}$$

- b. What is the inverse operation used to solve the equation $x + 23 = 78$?
- c. With your partner, discuss how to confirm whether or not the solution is accurate.



14.3.5 Your Turn!

1. The algebra tiles below model an equation.



- a. How many blue tiles are equivalent to one purple tile?
- b. Write a one-variable equation to represent the tiles on the mat.
- c. What inverse operation can be used to solve the equation?
- d. What value of y makes the equation true?

2. The equation $-31 = s - 6$ can be solved using inverse operations.

a. Fill in the missing information to solve for s .

$$-31 = s - 6$$

$$-31 + \underline{\hspace{2cm}} = s - 6 + \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} = s$$

b. Which inverse operation was used to solve the equation?

c. Check your solution.

3. Two bar model diagrams are shown.

Bar Model Diagram 1	Bar Model Diagram 2								
<table><tr><td colspan="2">m</td></tr><tr><td>-12</td><td>-4</td></tr></table>	m		-12	-4	<table><tr><td colspan="2">-12</td></tr><tr><td>4</td><td>m</td></tr></table>	-12		4	m
m									
-12	-4								
-12									
4	m								

a. Write an algebraic equation for each diagram box diagram.

Algebraic Equation for Diagram 1	Algebraic Equation for Diagram 2

b. Find the value of variable, m , for each algebraic equation.

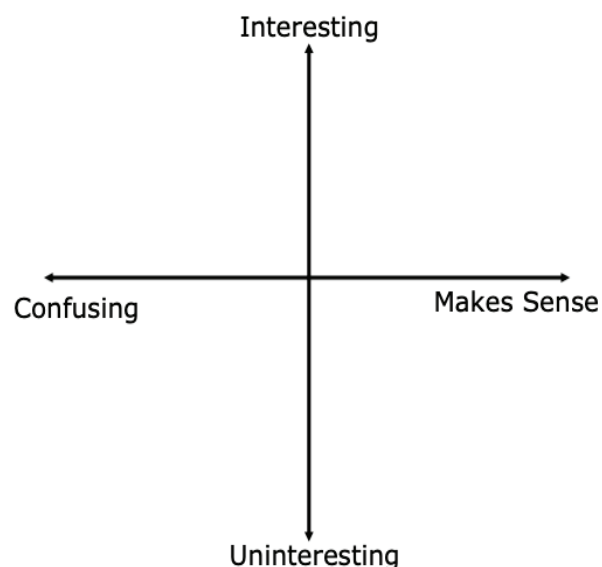
Diagram 1	Diagram 2

c. Discuss with your partner what you notice about the two equations. Summarize your discussion.



14.3.6 Wrap-Up: Reflection

1. Plot a point to indicate how you feel about today's learning target: "I can use diagrams and inverse operations when solving one-step addition and subtraction equations."



Unit 14, Lesson 4: Solving One-Step Equations in One Variable Using Addition and Subtraction



Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



Learning Target

- ☐ I can solve addition and subtraction one-step equations.



14.4.1 Warm-Up: Visualizing Solutions

In the two previous lessons, visual representations such as number lines, bar models, hanger diagrams, balance scales, and drawings were used to represent equations.

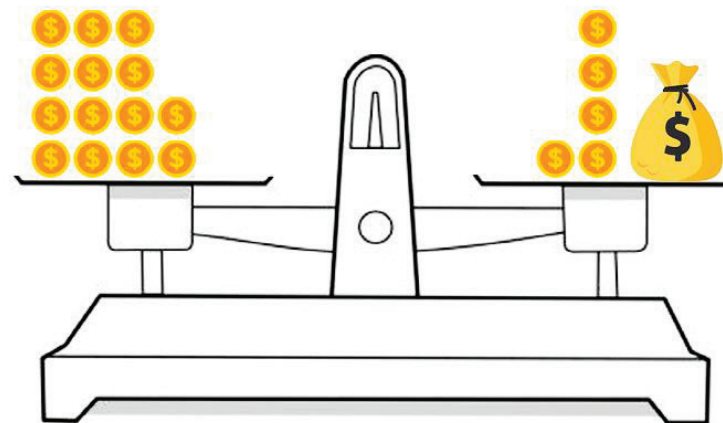
1. Use the visual representation of your choice to represent the equation $37 = n + 12$.

2. Find the value of n .



14.4.2 Collaboration: Algebraic Equations from a Diagram

1. Ellen and Frey were given the balance scale shown below, and told to write and solve an algebraic equation using inverse operations.



Ellen said, "There's no way to write an algebraic equation. If we can write an algebraic equation, we do not need inverse operations to solve it."

Frey said, "The only way to solve an algebraic equation is to use inverse operations to show the math is being reversed."

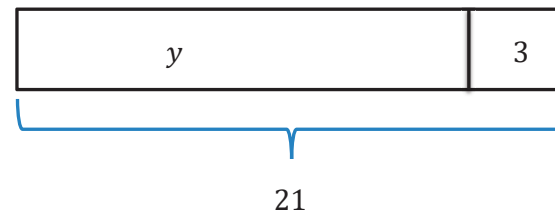
- a. Whose statement do you agree with and why? Explain your answer to your partner.

b. Write the algebraic equation represented by the balance scale.

c. Solve the equation and show the steps you took.

d. How many gold coins are in the bag? Compare your answer with your partner.

2. A bar model is shown.



a. Write an addition equation and a subtraction equation that can be represented by the bar model.

Addition Equation	Subtraction Equation

b. Describe the relationship between the two equations.

c. Find the value of y using either equation.

d. Check your solution with your partner. Make any revisions if necessary.



14.4.3 Guided Practice: Solving One-Step Algebraic Equations Using Inverse Operations

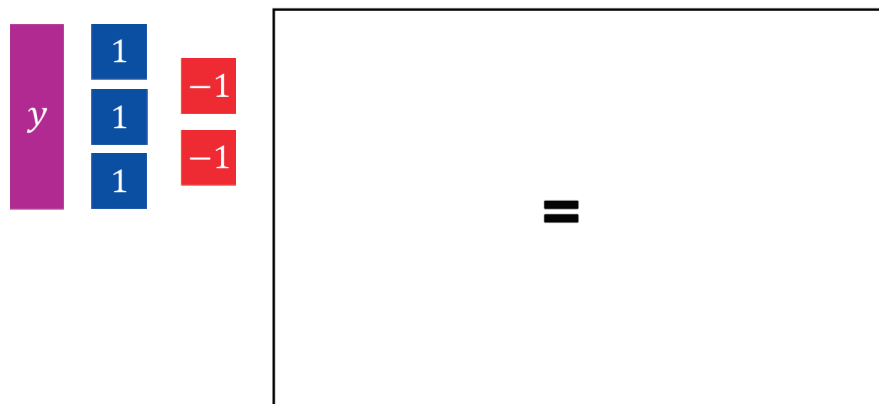
1. Use the equation $31 = m - 19$ to answer the following.
 - a. What operation is being performed in the equation?
 - b. What is the inverse operation?
 - c. Solve the equation using the inverse operation to find the value of m . Show your work.
 - d. Verify that the solution is correct by substituting the value found for m into the original equation.
2. Use the equation $d + 64 = -12$ to answer the following:
 - a. What is the inverse operation that will be used to solve for d ?
 - b. Write and solve the equation using the inverse operation.
 - c. What is the value of d ?
 - d. Check your solution with your partner.



14.4.4 Exploration: Solving Equations Using Opposites

Work with your partner to complete the following.

1. Arrange the algebra tiles provided on the equation mat so that they represent the equation $-2 + y = 3$.

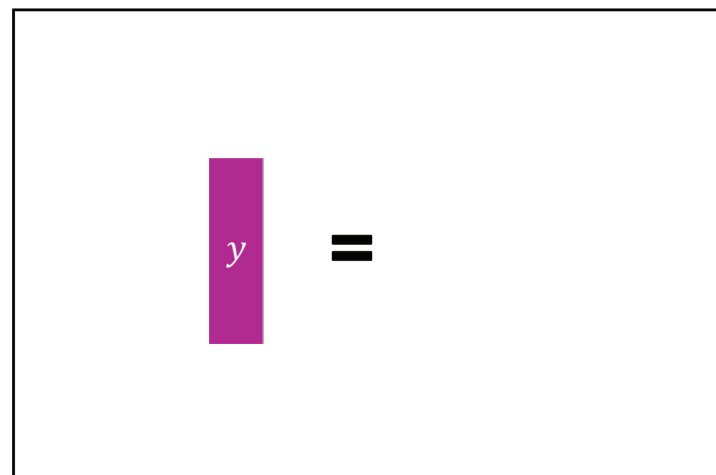


2. Check your diagram with your partner.
3. For this equation, there are no tiles that can be removed from both sides of the equation to find the value of y . Discuss with your partner how zero pairs could be used to find the value of y .

4. The goal when solving an equation is to isolate the variable. Which tiles can be added to both sides of the equation so that all tiles other than y on the left side of the equation can be eliminated?

5. Explain why, when tiles are added to one side of the equation, they also must be added to the other side.

6. Draw tiles on the equation mat to represent the solution to the equation.



7. What is the value of y ?

8. Discuss with your partner how solving for the variable in the equation $-2 + y = 3$, was different from solving for the variable in the equation $d + 64 = -12$, even though both equations include addition. Summarize your discussion.



14.4.5 Collaboration: Solving Equations

With your partner, determine who will be Partner A and who will be Partner B.

1. Solve the equations from your designated column.

Partner A	Partner B
$18 = 8 + x$	$-8 = -18 + x$
$p - 9 = 16$	$9 = p - 16$
$469 = 456 + w$	$-469 + w = -456$

2. The equations in each row have the same solutions. Check your work with your partner, making revisions to your work if necessary.



14.4.6 Your Turn!

1. Solve each equation. Show your steps in finding the solution.

a. $h + 37 = 62$

b. $18 = y - 23$



14.4.7 Wrap-Up: Cool Down

1. Your friend Giselle is absent today and will need the notes from class. Write a quick note to help her understand how to use inverse operations to solve the equation $25 = y + 7$.

Unit 14, Lesson 5: Writing and Solving One-Step Addition and Subtraction Equations Within Real-World Contexts



Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



Learning Target

- ☐ I can write and solve one-step equations in one variable within a real-world context using addition and subtraction.



14.5.1 Warm-Up: Bell Work

1. Before last night, Faith had p Instagram posts on her account. She made 4 more posts last night. She now has 313 Instagram posts on her account. Write an equation that can be used to find p , the number of posts Faith had on her account before last night.



14.5.2 Collaboration: Solving Equations in Real-World Contexts

Work with your partner to complete the following.

1. Kiran's cat weighs 5 pounds less than Clare's puppy. Clare's puppy weighs 14 pounds.
 - a. Select all of the equations that could be used to find the weight of Kiran's cat.

- ☐ $x + 5 = 14$
 - ☐ $x - 5 = 14$
 - ☐ $x = 14 + 5$
 - ☐ $x = 14 - 5$
 - ☐ $x - 14 = 5$
 - ☐ $x + 14 = 5$
 - b. What does the variable, x , represent in this situation?
 - c. What is the weight of Kiran's cat?

2. Kiegan collected 7 more items than his cousin Joel did for the annual toy drive. Joel collected 12 items.
 - a. Select all of the equations that could be used to find the number of items Kiegan collected.

- ☐ $t = 7 - 12$
 - ☐ $t - 7 = 12$
 - ☐ $t + 7 = 12$
 - ☐ $t = 12 + 7$
 - ☐ $t + 12 = 7$
 - ☐ $t - 12 = 7$
 - b. What does the variable, t , represent in this situation?
 - c. How many items did Kiegan collect for the toy drive?

Discuss with your partner what you notice about the equations that represent the two situations. Summarize your discussion.



14.5.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value

1. Jamal, a student at McLaughlin Middle School in Polk County, is preparing for a trip to Disney World with his mom and dad. An adult ticket is \$139. A child's ticket is \$5 less than an adult ticket. Jamal's parents are going to spend a total of \$412, before tax, on tickets.
 - a. Write an algebraic equation to represent the relationship between the cost of a child's ticket and the cost of an adult ticket in this situation.
 - b. What does the variable in your equation represent?
 - c. Solve the equation.
 - d. What was the cost of Jamal's ticket?
2. After doing some additional research, Jamal's mom also decided to purchase the meal bundle for their trip. The meal plan price is different for adults than children. It costs \$110 for his mom and dad. The total for all three meal plans was \$136. An equation can be written to find the unknown cost of Jamal's meal plan.
 - a. Write an algebraic equation that can be used to find the cost of Jamal's meal plan.
 - b. What does the variable in your equation represent?
 - c. Solve the equation.
 - d. How much was Jamal's meal plan?

3. Use the equation $49 + t = 107$ to complete the following.

- Write a word problem that can be represented by the equation $49 + t = 107$.
- What does the variable represent based on the word problem you created?
- What is the value of the variable?
- Share **only** your word problem with your partner. Use your partner's word problem to fill in the table.

Partner's Word Problem	What Does the Variable Represent?	Solve the Equation	Partner's Initials



14.5.4 Collaboration: Writing and Solving Equations for Real-World Contexts

Work with your partner to complete the following.

- PJ and Jasmine were given the following word problem.

Lexi's school is having a food drive. At the end of today's lunch period, Lexi collected a total of 34 non-perishable items for the food drive. Lexi celebrated, as it was 11 more than yesterday's lunch collection! How many total non-perishable items were collected yesterday?

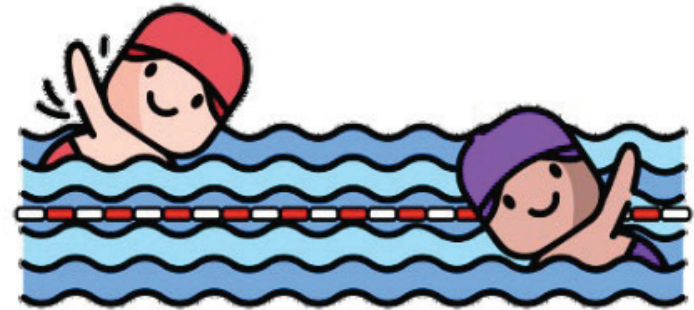
Jasmine told PJ they needed to solve the problem using an addition equation. PJ disagreed and said they need to use a subtraction equation.

- With your partner, determine who will follow Jasmine's suggestion and who will follow PJ's. Then, write the equation based on whose suggestion you are assigned and solve.

Jasmine's Suggestion "Addition"	PJ's Suggestion "Subtraction"

- b. Compare your equations and discuss your solutions with your partner.
- c. Explain who was correct in their initial suggestion – Jasmine, PJ, both of them, or neither of them.
- d. How many non-perishable food items were collected during Lexi's lunch period yesterday?

2. An image of two swimmers in a pool is shown.



- a. Work with your partner to create an addition or subtraction word problem based on the image.
- b. Write an equation to represent your scenario.
- c. What does the variable represent in your equation represent?
- d. What is the value of the variable?



14.5.5 Your Turn!

Write and solve an algebraic equation for each of the following.

1. Priya and Gerri are working on an extra-credit poster for math class. Priya has 12 double-tip markers, which is 3 fewer than Gerri has. How many markers does Gerri have?
2. The most popular elective at Arts Academy is the improv class. Out of the 97 students that have applied, only 45 can be accepted. How many students are not accepted?



14.5.6 Wrap-Up: Reflection

1. Shade in the arrow up to the statement that best describes your current understanding of the benchmark below:

Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



I'm so confident, I could explain how to write and solve one-step equations in one variable using addition or subtraction to someone else!

I can get to the right answer, but I don't understand well enough to explain how to write and solve one-step equations in one variable yet.

I understand some of this, but I don't understand all of it yet.

I have been trying hard, but I am finding this challenging.

Unit 14, Lesson 6: Solving One-Step Multiplication and Division Equations in One Variable Using Models



Benchmark

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.



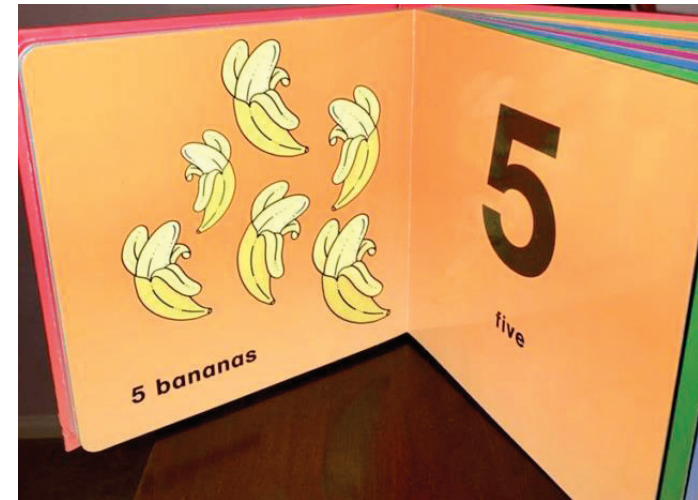
Learning Targets

- ☐ I can solve one-step multiplication and division equations using manipulatives.
- ☐ I can solve one-step multiplication and division equations using diagrams.



14.6.1 Warm-Up: Math Fail

This children's book has published a "math fail."



1. Explain the error shown in this image.



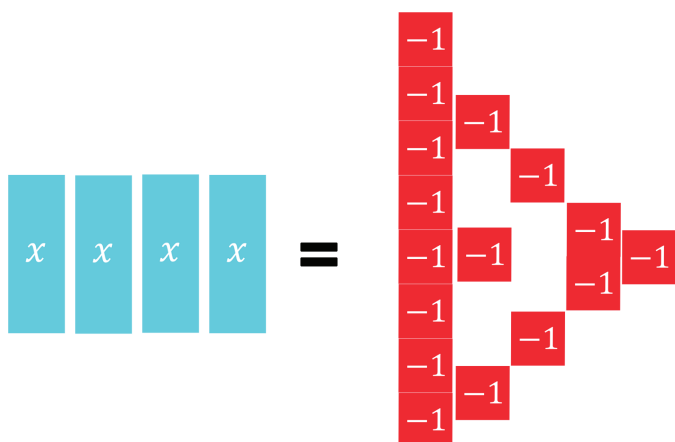
14.6.2 Exploration: Exploring Relationships

Recall that when solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. A model of the equation $4x = -16$ is shown using algebra tiles.

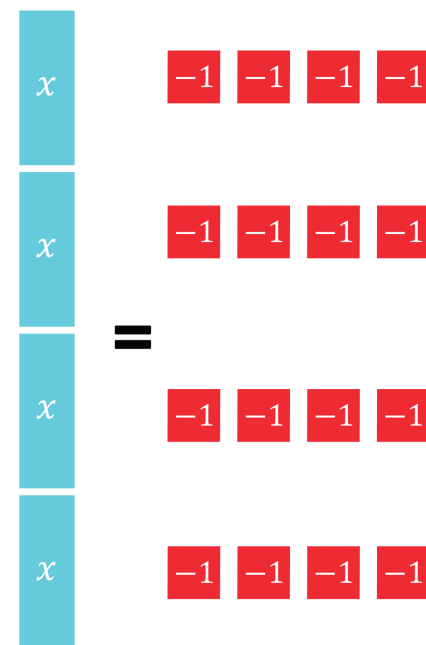


The  tile represents the variable, x , and the  represents the constant term where each tile is 1 unit.

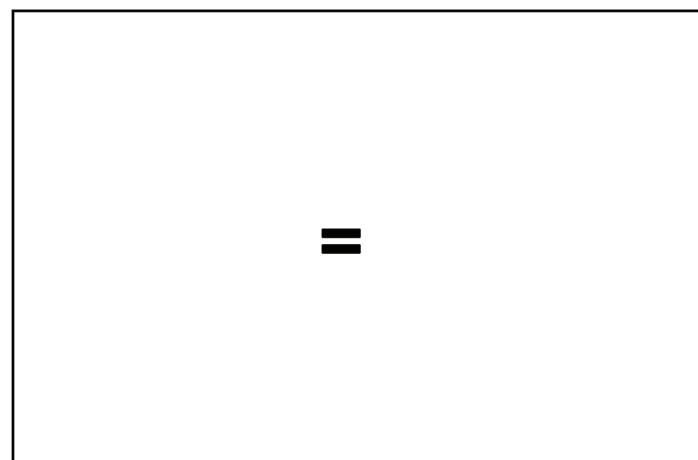


- a. What operation is being applied to the 4 and the x in the equation $4x = -16$?

- b. The algebra tiles are arranged on the equation mat into equal-sized groups on both sides of the equation. How many groups were created?

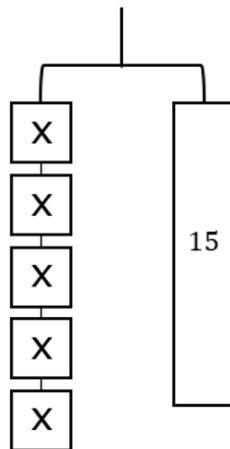


- c. Use algebra tiles to sketch the solution of the equation.



- d. What is the value of x ?

2. The hanger diagram represents the equation $5x = 15$.



- a. Complete the statements.

The left side of the hanger is broken into
equal parts. If 15 is broken into the same number of
equal parts, each part will have the same weight as

- ☐ 3
- ☐ 5
- ☐ 15

one x block. Splitting each side of the hanger into

- ☐ 3
- ☐ 5
- ☐ 15

the same as

- ☐ dividing
- ☐ multiplying

each

side of the equation by

- ☐ 3.
- ☐ 5.
- ☐ 15.

- b. Sketch splitting the hanger diagram into the equal parts as described in Part A.

- c. What is the value of x ?

- d. Compare your diagram with your partner. Discuss the similarities and differences.

- e. Provide positive feedback to your partner in their book on their sketch and solution.

Partner Feedback	
-------------------------	--

3. Inverse operations can be helpful in solving equations.

- a. Draw an arrow to match each operation on the left with its inverse operation on the right.

division

division

addition

addition

subtraction

subtraction

multiplication

multiplication

- b. Explain what it means for operations to be inverses.

- c. Which inverse operations were used to solve the equations in this exploration?



14.6.3 Guided Instruction: Using Models to Solve Equations

Algebra tiles can also be used to model division equations.

1. The equation $\frac{x}{3} = 4$ can be rewritten as $x \div 3 = 4$.

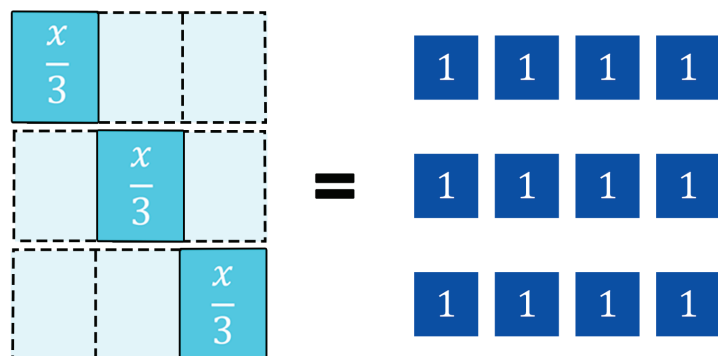
To represent division, show that x tile is being divided into equal parts. Sketch lines on the x tile to represent the fraction $x \div 3$.

This means when x is divided into 3 parts, each part is equal to 4. The model shown illustrates that $\frac{x}{3} = 4$.

$$\frac{x}{3} = 1 + 1 + 1 + 1$$

- a. If each $\frac{1}{3}$ of x is equal to 4, discuss with your partner what the value of x is. As part of your discussion, include which operation can be used to determine the value of x . Summarize your discussion.

- b. The image models the relationship discussed that can be used to determine the value of x .



Explain how this representation can be used to determine the value of x .

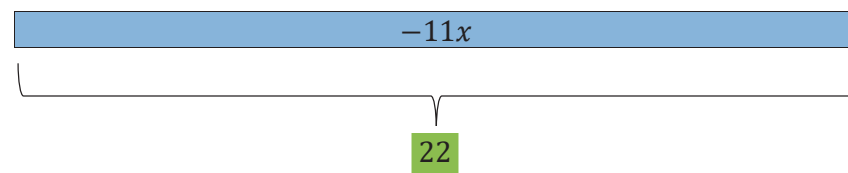
- c. What is the value of x ?

- d. Complete the statements by writing operations on each blank.

The original equation involved the operation _____. Its inverse operation, _____, was used to solve the equation.

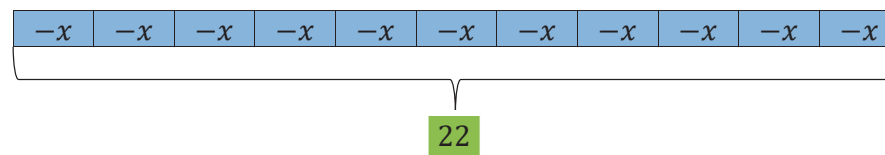
Tape diagrams, also known as bar models, can be used to model and solve multiplication and division equations.

2. The tape diagram models the equation $22 = -11x$.



The solution to the equation $22 = -11x$ can be modeled in the following steps. Work with a different partner to explain what is occurring in each step.

Step 1



- a. How were the integers in the equation used to determine the negative value of x ?

Explanation of Step 1

Step 2

$$\frac{-x}{2}$$

Explanation of Step 2

Step 3

$$\frac{x}{-2}$$

Explanation of Step 3

- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.



14.6.4 Collaboration: Matching Equations and Solutions

1. Match each equation to the correct solution. Record the equation with its solution in the table. Some solutions may be used more than once or not at all. Include an explanation for the match.

A. $x = 2$	E. $x = 3$
B. $x = -20$	F. $x = 10$
C. $x = -10$	G. $x = -8$
D. $x = 9$	H. $x = -3$

1. $-3x = -6$

2. $-\frac{x}{2} = 4$

3. $\frac{x}{3} = 3$

4. $-\frac{x}{2} = 5$

5. $3x = 9$

6. $\frac{x}{10} = -2$

7. $4x = 8$

Equation	Solution	Explanation for Match



14.6.5 Wrap-Up: Ticket Out the Door

- Throughout this unit, you have been shown how to solve equations using tape diagrams, hanger diagrams, number lines, manipulatives, and inverse operations. Use one of these methods to model and solve the equation $6h = -12$.

Unit 14, Lesson 7: Solving One-Step Multiplication and Division Equations Algebraically



Benchmark

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.



Learning Target

- ☐ I can solve one-step multiplication and division equations in one-variable algebraically.



14.7.1 Warm-Up: Different Problems, Same Solution

1. Write four equations whose solutions are all $x = 3$.



14.7.2 Guided Instruction: Using Inverses to Solve Equations

Inverse operations are used to solve equations by isolating a variable of interest.

When solving one-step equations using visual models, the inverse operation performed on one side of the model is also performed on the other side to keep the equation balanced.

1. With your partner, re-visit the meaning of balanced equations. Write a brief summary explaining what it means for equations to be balanced and how balance is maintained when solving. Include examples as part of your summary.
2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Multiplication ($\times$)	
Division ($\div$)	

3. Consider the equation $7m = 21$ to complete the following.
 - a. Discuss with your partner how to solve $7m = 21$.
 - b. What operation is occurring between the variable m and its coefficient of 7?
 - c. To isolate the variable m from its coefficient of 7, what inverse operation should be performed?
 - d. Work with your partner to determine the value of m .

4. Consider the equation $13 = \frac{h}{-5}$ to complete the following.

a. What mathematical operation is represented by $\frac{h}{-5}$?

b. What is the inverse operation?

c. Perform the inverse operation in the boxes below.

$$\boxed{} (13) = \left(\frac{h}{-5}\right) \boxed{}$$

d. What is the value of h ?

e. Discuss with your partner how to check whether or not your solution is correct. Then, check your solution.



14.7.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically

1. Complete the table.

Equation	Description of How to Solve	Work	Solution
$6x = 18$	Divide both sides of the equation by 6.		
$\frac{h}{5} = 7$			
$-36 = \frac{m}{-6}$			
$-32 = 8y$			
$4 = -\frac{b}{12}$			

2. With your partner, verify each solution is correct using substitution.



14.7.4 Your Turn!

Solve each equation algebraically. Show your work.

1. $\frac{a}{4} = 10$

2. $-11 = \frac{x}{8}$

3. $-12p = -84$

4. $-42 = 6x$

5. $18c = 72$

6. $\frac{r}{6} = 36$



14.7.5 Wrap-Up: Error Analysis

1. While checking her homework from last night, Lola noticed she got the following problem incorrect. Examine Lola's work and write a note to her detailing how she can correct her mistake.

$$\begin{array}{r} 42 = h \div 7 \\ +7 \quad +7 \\ \hline 49 = h \end{array}$$

Unit 14, Lesson 8: Writing and Solving One-Step Equations in One Variable



Benchmarks

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.



Learning Targets

- ☐ I can write and solve one-step equations in one variable within a real-world context using multiplication and division.
- ☐ I can solve one-step equations in one variable using addition and subtraction.



14.8.1 Warm-Up: One-Step Equations

1. Fill in each blue box to make the equations true. Use only digits 1 to 9, at most one time each.

$$\boxed{} + a = \boxed{}$$

$$\boxed{} b = \boxed{}$$

$$c - \boxed{} = \boxed{}$$

$$a = \boxed{}, b = \boxed{}, c = \boxed{}$$



14.8.2 Collaboration: Relating Real-World Contexts to Equations

1. Four equations are shown.

$$4x = 32$$

$$32x = 4$$

$$\frac{x}{32} = 4$$

$$\frac{x}{4} = 32$$

Discuss with your partner which of the equations correctly represents each situation below, where x represents the unknown quantity. Then, write the matching equation in the space provided.

Situation 1	Situation 2
At a restaurant, 4 friends decided to split the bill evenly. If each person paid \$32, what was the total bill?	A manatee has a total of 32 teeth. If a manatee has 4 rows of teeth with the same number of teeth in each row, how many teeth are in each row?
Equation:	Equation:



14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division

1. A town's total budget for firefighters' wages is \$320,000. Wages are calculated at \$40,000 per firefighter.
 - a. Draw a bar model that can be used to determine how many firefighters the town can afford to employ.
 - b. Write an equation that can be used to find w , the number of firefighters the town can employ.
 - c. What is the value of w ?
 - d. Explain what the solution means in the context of the problem.

2. There are 4 flooring companies submitting bids on a new housing development. Each company has 11 different flooring samples.

Dino said, "To find the total number of samples, s , being submitted, set up the equation $s = 4 \cdot 11$." Dino's partner Callahan shared that he used the equation $\frac{s}{4} = 11$ to find the total number of samples submitted.

Bristol shouted across the classroom that she wrote the equation, $\frac{s}{11} = 4$.

- a. Discuss with your partner which student — Dino, Callahan, or Bristol — wrote the correct equation to solve for the total number of samples submitted.
- b. Solve one of the equations and then use a different equation to check your work.
- c. Explain how the answer makes sense in terms of the context.



14.8.4 Guided Practice: Solving Multiplication and Division Equations for Real-World Contexts

For each situation, write an equation that can be used to determine the unknown value. Then, solve the equation. Be sure to check your solution.

1. It takes 5 yards of fabric to make a Native American ceremonial dress.
 - a. Write a division equation and a multiplication equation that can be used to determine how many yards, d , of fabric are needed to make 17 ceremonial dresses.

Division Equation	Multiplication Equation

- b. Solve either of the equations for d . Use the other equation to check your work.

2. Drake is selling candy bars for a school fundraiser. Each candy bar sells for \$3.00.
 - a. Write a division equation and a multiplication equation that can be used to determine how many candy bars, c , he sold if he raised \$399.

Division Equation	Multiplication Equation

- b. Solve either of the equations for c . Use the other equation to check your work.

3. Discuss with your partner how creating the two different equations in Questions 1 and 2 demonstrate inverse relationships.



14.8.5 Try It: Mixed Review

Select all of the equations that represent each situation.

1. One Valencia orange tree can produce 275 oranges. A crate contains 25 oranges.

- a. How many crates can one Valencia orange tree fill up? Select all that apply.

☐ $\frac{x}{25} = 275$

☐ $\frac{x}{275} = 25$

☐ $275 = 25x$

☐ $\frac{275}{x} = 25$

☐ $275x = 25$

☐ $\frac{275}{25} = x$

- b. Pick one of the equations you selected and solve the problem. Show your reasoning.

2. Aliyah and Tiana want to bake 3 carrot cakes. Each carrot cake needs 2 cups of salted butter.

- a. How much butter will they need to bake all 3 cakes? Select all that apply.

☐ $\frac{x}{2} = 3$

☐ $6x = 2$

☐ $2x = 3$

☐ $\frac{x}{3} = 2$

☐ $2(3) = x$

☐ $\frac{x}{3} = 6$

☐ $3x = 2$

- b. Select one of the equations you selected and solve the problem. Show your reasoning.

3. Solve each equation for the variable. Verify your answers.

a. $5x = 230$	b. $y + 4 = -2$
c. $-16 + d = -26$	d. $r - 96 = 103$
e. $\frac{k}{4} = -51$	f. $-156 = -13w$



14.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning objective.



I need help.



I got it.



I could help someone else.

Determine the value of an unknown decimal or fraction in an equation.	  
Translate a written description into an algebraic expression.	  
Write a description within a real-world context given an algebraic expression.	  
Write and solve one-step equations in one variable using addition and subtraction.	  
Write and solve one-step equations in one variable using multiplication and division.	  

Unit 14, Lesson 9: Solving One-Step Inequalities Using Addition and Subtraction



Benchmark

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.



Learning Targets

- ☐ I can relate the solution set of an inequality to its graph on a number line.
- ☐ I can relate the solution set of an inequality to the context it represents.
- ☐ I can determine the minimum or maximum value that makes an inequality true.
- ☐ I can solve one-step inequalities in one variable using addition or subtraction.
- ☐ I can relate properties of inequality to solving one-step inequalities algebraically.



14.9.1 Warm-Up: Which One Doesn't Belong?

Four inequalities are shown.

Inequality A	Inequality B	Inequality C	Inequality D
$x \geq 5$	$12 \leq 7$	$2 > x$	$x > 1$

1. Javier says that Inequality B does not belong with the others. What is one reason Javier might have said this?

2. Select a different inequality and explain why it doesn't belong.



14.9.2 Exploration: Representing Inequalities in Context

Height restrictions are often in place for roller coasters and rides at amusement parks to keep guests safe. At Disney's Typhoon Lagoon Water Park, the following height restrictions are in place.

Attraction	Height Restriction
Bay Slides	Guests must be 60 inches or shorter
Crush 'n' Gusher	Guests must be 48 inches or taller
Ketchakiddee Creek	Guests must be 48 inches or shorter

1. Work with your partner to determine which graph represents the height restrictions of each attraction at Typhoon Lagoon. Then, draw an arrow from each attraction to its matching graph.

Attraction	Graph Representing Safe Heights for the Attraction
Bay Slides	
Crush 'n' Gusher	
Ketchakiddee Creek	

2. Nick is happy to realize that he is tall enough to go on the Crush 'n' Gusher raft ride. Nick's friend Matt is 2 inches shorter than Nick. Explain whether Matt is tall enough to go on the ride with Nick.

3. Complete the statements.

If Nick is n inches tall, then Matt is ☐ $n + 2$
☐ $n - 2$ inches tall.

For Matt to be tall enough to go on Crush 'n' Gusher,

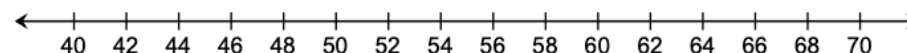
the value of ☐ $n + 2$
☐ $n - 2$ must be ☐ less than
☐ greater than or

equal to ☐ 48.
☐ 60. Therefore, the solutions to the

inequality ☐ $n + 2 \leq$
☐ $n + 2 \geq$
☐ $n - 2 \leq$
☐ $n - 2 \geq$ ☐ 48
☐ 60 represent Nick's height

that mean Matt could also ride the Crush 'n' Gusher.

4. On the number line, show all the possible heights that Nick could be if Matt is also tall enough to go on the ride.



5. What is the minimum height Nick must be in order for Matt to be able to go on the ride as well?

Julia noticed the graph also represents the solutions to the inequality $n \geq 50$.

6. Work with your partner to explain why the solutions to the inequality $n - 2 \geq 48$ can be represented the same way on a number line as $n \geq 50$.



14.9.3 Bring It Together: Inequalities in Mathematical and Real-World Contexts

Inequalities have an infinite number of solutions. The set of all the solutions of an inequality is called the **solution set**.

The **solution set** of an inequality includes the values that, when substituted for the variable, make the inequality true.

One way to represent the solution set of an inequality is by shading the values on the number line, beginning with the minimum or maximum value, and including all the values to the right or left of it that are included in the solution set.

1. Complete each statement.

A minimum or maximum value represented by an open circle means the value is ☐ included in ☐ excluded from the solution set.

A minimum or maximum value represented by a closed circle means the value is ☐ included in ☐ excluded from the solution set.

Graphs representing the solution set of inequalities are helpful when interpreting them in real-world contexts. However, sometimes there are parts of the solution set that may not make sense in a given context.

For example, to ride the Bay Slides at Typhoon Lagoon, guests must be 60 inches or shorter. The graph below represents the graph of $h \leq 60$ in this situation, where h is the height of guests that can ride the Bay Slides.



2. What does 60 represent in this situation?
3. What does the arrow pointing to the left on the number line represent?
4. Describe at least two values that are part of the solution set of $h \leq 60$ that do not make sense in this context.



Sometimes there are values that are part of the mathematical solution set that do not make sense in the context. In those cases, the variable may have a **constraint** (restriction or limitation) placed on it so that the mathematical model is realistic.

The Exploration demonstrated the inequalities $n - 2 \geq 48$ and $n \geq 50$ are equivalent because all of the same values of n make both inequalities true. Both inequalities were also represented using the same graph.

Inequalities that have the same solution set are called equivalent inequalities.



14.9.4 Exploration: Properties of Inequality

Three inequalities are shown.

$$x > -2.3 \qquad -2.3 > x \qquad -2.3 < x$$

1. Circle the inequalities that represent the same solution set.

2. Complete the statement.

For both inequalities, all the values of x that are _____ than _____ make both inequalities true.

These two inequalities are equivalent by the **asymmetric property of inequality**.



The **asymmetric property of inequality** states that if $a > b$, then $b < a$.

3. Use the asymmetric property of inequality to write an inequality equivalent to $q + \frac{2}{5} \leq \frac{7}{10}$.

4. Four true inequalities are given in the table. Work with your partner to complete the table by following the steps indicated for each inequality and then answering the questions.

Given Inequality	Step	What is the new inequality?	Is the new inequality true?
$-6 < -2$	Add 8.1 to both sides		
$-5 < \frac{3}{4}$	Subtract 2 from both sides		
$7.45 > 7$	Add $3\frac{1}{5}$ to both sides		
$0 > -\frac{2}{3}$	Subtract $\frac{5}{6}$ from both sides		

5. Use your work on the previous problem to complete the description of each property.

Property of Inequality	Description
Addition property of inequality	When the same value is added to both sides of a true inequality, ...
Subtraction property of inequality	When the same value is subtracted from both sides of a true inequality, ...



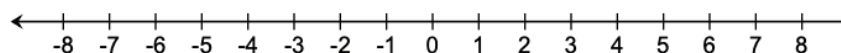
14.9.5 Guided Instruction: Solving One-Step Inequalities with Addition and Subtraction

1. Complete the table to determine the value of $x - 3$ for each value of x .

x	-4	-3.2	-2	$-\frac{1}{2}$	0	1	2	$3\frac{6}{7}$	4
$x - 3$	-7		-5				-1		1

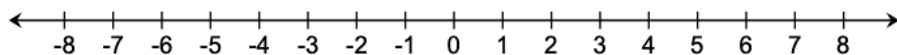
2. For which value of x is the equation $x - 3 = -2$ true?

3. Represent the solution to the equation $x - 3 = -2$ on the number line.



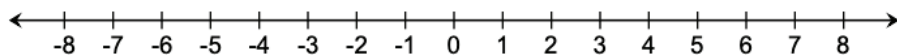
4. For which values of x in the table is the inequality $x - 3 > -2$ true?

5. Represent the values from the table that are part of the solution set of $x - 3 > -2$ on the number line.



The solution set of $x - 3 > -2$ includes more values than shown on the previous number line.

6. Graph the solution set for the inequality.



7. Write a value on the line to show an equivalent inequality represented by the graph above.

$$x > \underline{\hspace{1cm}}$$

The solution set of the inequality $x - 3 > -2$ is all the values of x that are greater than 1, or $x > 1$.

This solution set can be determined algebraically by adding the same value to both sides of the inequality.

8. What value can be added to both sides of the inequality $x - 3 > -2$ to isolate x ?

Adding the same value to both sides of an inequality can be used to solve an inequality by the ***addition property of inequality***.



Addition Property of Inequality

If $a > b$, then $a + c > b + c$.

The ***addition property of inequality*** states that if the same value is added to both sides of an inequality, the result will be an equivalent inequality that is also true.

The ***subtraction property of inequality*** can also be used to solve an inequality.



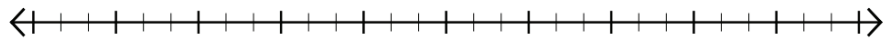
Subtraction Property of Inequality

If $a > b$, then $a - c > b - c$.

The ***subtraction property of inequality*** states that if the same value is subtracted from both sides of an inequality, the result will be an _____ inequality that is also _____.

These properties can be used to solve one-step inequalities algebraically.

9. Use the inequality $-12.25 > y - 2.4$ to complete the following.
- a. Find the solution set by isolating the variable. Show your work.
- b. Graph the solution set on the number line.
- c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.



14.9.6 Wrap-Up: Cool Down

1. Complete the table.

The terms and information used in today's lesson that I knew were...	Something I wondered during today's lesson was...	Something I learned during today's lesson was...



Unit 14, Lesson 10: Solving One-Step Inequalities Using Multiplication and Division



Benchmark

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.



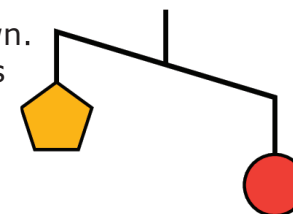
Learning Targets

- ☐ I can determine the minimum or maximum value that makes an inequality true.
- ☐ I can explore the relationship between the coefficients of the terms in an inequality and solution set.
- ☐ I can solve one-step inequalities in one variable using multiplication or division.
- ☐ I can relate properties of inequality to solving one-step inequalities algebraically.



14.10.1 Warm-Up: Visual Models

An unbalanced hanger diagram is shown. It shows that the weight of one circle is greater than the weight of one pentagon.



1. Complete the table.

Hanger Diagram	Description	Inequality
	The weight of one circle, c , is greater than the weight of one pentagon, p .	
		$s < p$
	The weight of one square, s , is _____ than the weight of a circle and a pentagon.	$c + p > \underline{\hspace{1cm}}$
		$s + c > 2t$



14.10.2 Exploration: Determining the Minimum or Maximum Value that Makes an Inequality True

Work with your partner to complete the following.

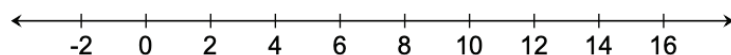
1. Bak Middle School of the Arts in West Palm Beach, Florida is hosting a rehearsal for its upcoming musical. The stage manager has \$83 from a recent fundraiser that can be used to order lunch for the cast and crew. The school's cafeteria can make boxed lunches for a cost of \$6 each. The inequality $6n \leq 83$ represents the situation, where n represents the number of boxed lunches that can be ordered from the cafeteria.
 - a. Select the values that represent a valid number of boxed lunches the stage manager can order from the cafeteria.
 - ☐ 14
 - ☐ $13\frac{5}{6}$
 - ☐ 12
 - ☐ 9.5
 - ☐ 2
 - ☐ -4
 - b. Explain why some values in the previous question are not valid in this context.

- c. The mathematical solution to the inequality $6n \leq 83$ is $n \leq 13\frac{5}{6}$. What is the maximum number of boxed lunches the stage manager can order from the cafeteria?
- d. What is the minimum number of boxed lunches the stage manager can order from the cafeteria?

- e. The graph of the solution set for $6n \leq 83$ is shown.



Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line shown.



2. Hiromi walks around his neighborhood each afternoon, tracking his pace each day. The fastest rate Hiromi can walk for an extended period of time is 3.7 miles per hour.
- a. What is the maximum distance Hiromi can walk in 0.75 hours?

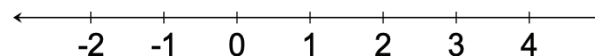
- b. The inequality $\frac{d}{3.7} \leq 0.75$ can be used to represent the distance Hiromi walks, d , in 0.75 hours given his fastest walking rate of 3.7 miles per hour.

Use your answer to Part A to determine the solution set for $\frac{d}{3.7} \leq 0.75$.

- c. The graph of the solution set for $\frac{d}{3.7} \leq 0.75$ is shown.



Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line.





14.10.3 Bring It Together: Solutions and Mathematical Models in Context

Each of the inequalities in the Exploration has a mathematical solution set that can be represented on a number line. These graphs are examples of mathematical models.



Mathematical models are helpful tools used to understand a situation and make predictions. However, sometimes the mathematical model can include parts that do not make sense in a real-world context. When interpreting models to solve problems, it is important to analyze whether aspects of the model make sense in a given context.



14.10.4 Collaboration: The Effect of a Negative Coefficient

1. Two simple inequalities are shown.

$$1 > 0 \quad \text{and} \quad -1 > 0$$

- a. Circle the inequality that is true and ~~cross out~~ the inequality that is false.

- b. Complete the statement.

If 1 is greater than 0, then the opposite of 1 is

_____ than 0.

- c. Write the correct inequalities by filling in a comparison symbol on each line.

$$0 \text{ ____ } 1 \quad \text{and} \quad 0 \text{ ____ } -1$$

- d. Verify your answers with your partner on parts A-C and, if necessary, resolve any differences.

2. Work with your partner to complete the following.

- a. Complete the inequalities in the table based on your understanding of opposites.

Given Inequality	Equivalent Inequality
$0 < -q$	$0 \underline{\hspace{1cm}} q$
$0 > -m$	$0 \underline{\hspace{1cm}} m$
$-w \leq 0$	$w \underline{\hspace{1cm}} 0$

In each equivalent inequality, the sign of the variable changed to its opposite and the value 0 remained the same.

- b. Describe which operation(s) could be applied to both sides of the given inequalities above to result in the signs of the variables changing and the value 0 remaining the same.
- c. Describe what happens to the comparison symbol in the inequality when either of these operations are applied.



14.10.5 Bring It Together: The Effect of a Negative Coefficient

1. Complete the statements using words from the bank shown. Some words may not be used.

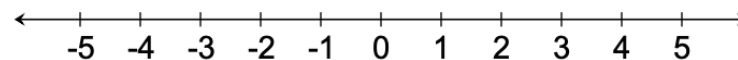
adding subtracting multiplying
dividing opposite

When the variable term in an inequality is negative, the inequality can be rewritten by _____ or _____ the inequality by -1 . In the resulting inequality, each term and the comparison symbol will change to its _____.

2. Use the inequality $-n \geq 2$ to complete the following.
- a. Complete the inequality to create an equivalent inequality.

$n \boxed{\hspace{1cm}} \underline{\hspace{1cm}}$

- b. Graph the solution set for $-n \geq 2$.





14.10.6 Guided Instruction: Solving One-Step Inequalities with Multiplication and Division

As with solving one-step inequalities with addition and subtraction, properties of inequality can be used to solve inequalities with multiplication and division. As discovered in the Collaboration, though, when the value that is being multiplied or divided is negative, this changes both the values in the inequality *and* the comparison symbol to their opposites.

Multiplying both sides of an inequality by the same value can be used to solve an inequality by the **multiplication property of inequality**.



Multiplication Property of Inequality

If $a > b$ and $c > 0$, then $a \times c \underline{\hspace{1cm}} b \times c$.

If $a > b$ and $c < 0$, then $a \times c \underline{\hspace{1cm}} b \times c$.

- Find the solution to both inequalities using the multiplication property of inequality.

$$\frac{y}{4} \leq 2.1$$

$$\frac{y}{-4} \leq 2.1$$

Dividing both sides of an inequality by the same value can be used to solve an inequality by the **division property of inequality**.



Division Property of Inequality

If $a > b$ and $c > 0$, then $a \div c \underline{\hspace{1cm}} b \div c$.

If $a > b$ and $c < 0$, then $a \div c \underline{\hspace{1cm}} b \div c$.

- Find the solution to both inequalities using the division property of inequality.

$$-25 > 5c$$

$$-25 > -5c$$



14.10.7 Wrap-Up: Cool Down

Two inequalities are shown.

$$2.5x \geq -10$$

$$-2.5x \geq -10$$

1. Complete the table to describe how the process of solving each inequality is similar and different.

How is the process of solving each inequality similar?	How is the process of solving each inequality different?

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